

# Sampling Design for Spatial Cluster Randomized Trials Under Heterogeneous Incidence

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## Introduction

Cluster randomized trials (CRTs) randomize intact groups, such as clinics, schools, or geographic areas, rather than individuals, and are the standard design when an intervention is delivered at the group level or when individual randomization would create unacceptable contamination between arms [1, 2]. A central assumption underlying standard CRT analysis is the no-interference component of the stable unit treatment value assumption (SUTVA): a cluster’s outcome must not depend on which other clusters were assigned to treatment [3]. This assumption, borrowed from individually randomized trial theory, can fail once clusters are spatially arranged, and so can the weaker assumption of partial interference, under which outcomes may depend on assignments within prespecified groups but not across them [4]. Tobler’s first law of geography, “everything is related to everything else, but near things are more related than distant things” [5], implies that an intervention delivered in one location can plausibly influence outcomes in geographically adjacent locations through shared populations, service networks, information diffusion, or environmental pathways. Documented examples of this kind of spatial spillover span public health and development economics: in a trial of insecticide-treated bed nets in northern Ghana, child mortality was lower in control villages close to treated villages than

in control villages farther away [6], and deworming interventions delivered in treated Kenyan schools reduced infection prevalence in nearby untreated schools through reduced transmission [7]. When control clusters benefit from nearby treatment in this way, an unadjusted comparison of treated and control clusters may understate the direct effect, and how much contamination occurs can depend on the spatial arrangement of treated and control clusters, not only on the number of clusters allocated to each arm.

## The design-stage gap in the spatial interference literature

The methodological literature addressing spatial interference in CRTs has concentrated on analysis-stage correction. Spatial autocorrelation statistics such as Moran’s  $I$  [8, 9] and Geary’s  $C$  [10] provide tools for detecting residual spatial dependence once outcomes have been observed, and spatial regression models, including spatial lag and spatial Durbin specifications [11], estimated by maximum likelihood [12] and implemented in modern software [13], allow an analyst to separate a direct treatment effect from a spillover effect after the fact. In their systematic review of spatial analysis methods for cluster randomized trials, Jarvis and colleagues placed trials that used spatial separation or buffers outside the primary review of statistical methods [14]. The literature therefore offers much more guidance on analyzing spatial effects than on how the choice of treatment assignment itself, made before any data are collected, might reduce the problem.

Design-stage tools do exist, but they were largely developed for a different purpose. Restricted randomization addresses the baseline imbalance that a limited number of clusters can produce [15]. Covariate-constrained randomization generates candidate assignments, retains those meeting a prespecified covariate-balance criterion, and randomizes among them [16]; Crisp and colleagues extended it to the selection of clusters from a larger eligible set and noted that their procedure also increases spatial dispersion among the selected clusters [17]. Crisp and colleagues’ dispersion criterion concerns which clusters are selected rather than how the selected clusters are assigned

to arms, and none of these methods targets the spatial pattern of arm assignment under spillover: balancing a covariate across arms does not by itself determine whether treated and control clusters are spatially interspersed in a way that induces contamination. A more recent body of work addresses the problem studied in this chapter more directly. Randomized saturation designs assign different saturations, the fractions of units treated, to different groups and randomize treatment within each group, which allows direct and spillover effects to be jointly identified rather than merely controlled for [18]. In the malaria control literature, a conventional “fried-egg” design omits each cluster’s outer area from analysis to limit contamination; McCann and colleagues proposed instead selecting whole clusters for inclusion or exclusion while preserving randomization [19]. Most recently, a cluster of theoretical papers published or posted between 2022 and 2025 has derived rate-optimal cluster-randomized designs under spatial interference [20], independent-set designs under network interference [21], designs that explicitly account for cross-cluster interference [22], and designs for spatio-temporally heterogeneous treatment effects [23]. This chapter complements this work with a simulation comparison of concrete assignment strategies that a trial team could implement, under heterogeneous baseline incidence and a range of spatial dependence and spillover conditions.

## **Extending a small-grid proof of concept**

This chapter extends the proof-of-concept study of Chapter 2, which asked a narrower version of the same question: how does a priori treatment-assignment strategy affect estimation under spatial spillover and spatial dependence? That study compared simple random assignment against a single block-stratified design, on small, exhaustively enumerable grids of 8 to 12 spatial units arranged in  $2 \times 4$ ,  $3 \times 3$ , and  $3 \times 4$  configurations, and estimated the intervention effect with a spatial autoregressive (SAR) lag model containing a spillover indicator, with separate parameters for a baseline effect, the intervention effect, the spillover effect, and spatial dependence. Among its future considerations, Chapter 2 listed a broader range of grid dimensions and neighbor

structures beyond rook contiguity, and it proposed comparing simple random assignment and block-stratified assignment with buffer-zone (“fried-egg”) designs, which exclude from analysis the control clusters that border treated clusters. This chapter takes up a larger grid and queen contiguity, and it adds two extensions: several further assignment designs, none of which excludes clusters from analysis, and heterogeneous, rather than spatially uniform, baseline outcome risk. Chapter 2 suggested weighting assignment by cluster population density; some of the designs here use baseline outcome incidence instead. We scale from 8–12 clusters to a 100-cluster grid, from two designs to six conceptually distinct treatment assignment strategies (two further variants, which could not be distinguished from a retained design, are reported in the appendix of this chapter), and from a single, uniform baseline risk surface to three distinct mechanisms of heterogeneous baseline incidence. This lets us ask which designs estimate the treatment effect well when baseline risk varies across space. Some of the designs use the incidence surface to assign treatment, and others ignore it.

## Aims

This chapter has three aims. First, to rank the six treatment assignment designs by estimation accuracy, measured by mean squared error (MSE), and by inferential validity, measured by 95% confidence interval coverage, separately within each of the three incidence-heterogeneity mechanisms, since a design that performs well under one mechanism may not generalize to another. Second, to test whether that ranking is stable across the spatial and spillover parameters that a real trial team cannot control or know in advance, including the magnitude of the true treatment effect itself, the strength of residual spatial dependence in the outcome, and the magnitude of the spillover effect. Third, to translate the resulting ranking into design guidance, illustrated with a planned sudden unexpected death (SUD) prevention pilot to be delivered through North Carolina’s community college network. The sections that follow describe the simulation methods, report results for each aim, present the SUD application, and discuss

100 limitations and future work.

## 101 **Methods**

### 102 **Spatial structure and outcome model**

103 We simulated a  $10 \times 10$  regular grid of 100 clusters under two definitions of adjacency. Under  
104 queen contiguity, two clusters are neighbors if they share an edge or a corner; under rook  
105 contiguity, they must share an edge. Queen contiguity is the primary analysis in this chapter.  
106 Rook contiguity is a sensitivity analysis and the case that continues Chapter 2, whose grids used  
107 rook neighbors. For each definition, the binary adjacency matrix was row-standardized to form  
108 the spatial weights matrix  $W$ , so that  $WZ$  gives, for each cluster, the proportion of its neighbors  
109 that are treated, and  $WY$  gives the average outcome among its neighbors. Row-standardization  
110 expresses each cluster’s spillover exposure as an average over its own neighborhood rather than  
111 a raw sum, which would otherwise depend on how many neighbors the cluster has. Within a  
112 scenario, the same neighbor type defines  $W$  in the outcome model, in the Isolation Buffer design,  
113 and in the fitted model.

114 Cluster-level outcomes  $Y$  were generated from a spatial lag model with a spillover term,

$$Y = (I - \rho W)^{-1} [\tau Z + S(Z) + \beta X + \epsilon], \quad \epsilon \sim N(0, \sigma^2 I), \quad (1)$$

115 where  $Z \in \{0, 1\}^{100}$  is the treatment assignment produced by the design under evaluation,  $\tau$   
116 is the direct treatment effect of interest,  $S(Z)$  is the spillover term defined in Equation 2,  $\rho$  is  
117 the spatial autoregressive coefficient governing dependence in the outcome beyond that induced  
118 by treatment and spillover,  $X$  is the baseline incidence covariate with coefficient  $\beta = 1$ , and  
119  $\sigma = 1$ . The spatial multiplier  $(I - \rho W)^{-1}$  propagates each cluster’s shocks through the network  
120 of neighbors. We tested  $\rho \in \{0, 0.01, 0.20, 0.50\}$ . When  $\rho = 0$  the multiplier is the identity and

the model is a linear regression with independent errors,  $Y = \tau Z + S(Z) + \beta X + \epsilon$ , in which spatial structure enters only through the spillover term; larger values of  $\rho$  induce progressively stronger spatial clustering in the outcome.

We evaluated two spillover regimes,

$$S(Z) = \gamma WZ \quad (\text{both arms}), \quad S(Z) = \gamma WZ \circ (1 - Z) \quad (\text{control only}), \quad (2)$$

where  $\circ$  denotes the elementwise product. Under the control-only regime, only untreated clusters respond to the treatment status of their neighbors, which represents classical contamination: an untreated cluster's outcome rises with the share of its neighbors that are treated. Under the both-arms regime, every cluster responds to its neighbors' treatment regardless of its own assignment, so neighboring treatment adds to a treated cluster's own effect. We tested four spillover magnitudes,  $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$ , and five magnitudes of the direct effect,  $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ , with  $\tau = 1$  as the primary scenario throughout.

## Heterogeneous baseline incidence

Chapter 2 assumed a spatially uniform baseline outcome risk. Cluster-level health outcomes typically vary across space, and the designs studied here differ in whether and how they use that variation. We generated the baseline incidence covariate  $X$ , with values in  $(0, 1)$ , under three mechanisms. Under the independent (iid) mechanism, each  $X_i$  is drawn from a  $\text{Uniform}(0, 1)$  distribution, with no spatial structure. Under the spatially autocorrelated mechanism,  $X = \Phi\{(I - \rho_X W_Q)^{-1}u\}$ , where  $u \sim N(0, I)$ ,  $\Phi$  is the standard normal distribution function applied elementwise,  $W_Q$  is the queen weights matrix, and  $\rho_X \in \{0.20, 0.50\}$ . Under the Poisson mechanism, a spatially autocorrelated log-rate surface,  $\log \lambda = \log(35/100,000) + (I - \rho_X W_Q)^{-1}u$  with  $\rho_X \in \{0.20, 0.50\}$ , generates cluster death counts  $C_i \sim \text{Poisson}(\lambda_i P)$ , with an equal population of  $P = 100,000$  in every cluster, so that a cluster at the base rate expects 35 deaths.

The observed rates  $C_i/P$  are then rank-normalized to  $X_i = (r_i - 0.5)/N$ , where  $r_i$  is the rank of cluster  $i$ 's rate among the  $N = 100$  clusters and tied rates share their average rank. Ties remain at this population size: the Poisson surfaces used here have between 60 and 76 distinct values among the 100 clusters. The base rate of 35 per 100,000 was chosen to be representative of published SUD incidence: it lies between the SUD incidence rates that Mirzaei and colleagues estimated for US men (49.3 per 100,000) and women (21.7 per 100,000) aged 20–64, close to their average [24]. It is not the observed rate of the county data described in the Application section, which come from a different source, age range, period, and case definition. The incidence filters always use the queen weights  $W_Q$ , whichever neighbor type the scenario uses, so the incidence surfaces are identical under rook and queen contiguity. The iid mechanism and the two strengths of each of the other two mechanisms give five incidence configurations.

For each configuration we generated  $K = 10$  incidence surfaces,  $X_1, \dots, X_{10}$ , each from its own seed. The same ten surfaces are used by every scenario in that configuration, across neighbor types, outcome-model parameters, designs, and values of  $\tau$ ; only the design draws and the outcome noise differ across those scenarios. Each design is applied to the surface that generates the outcome: assignments for surface  $k$  are drawn from  $X_k$ , and the outcome and the fitted model for those assignments use  $X_k$  as well. This assumes that the incidence observed when a trial is planned is the incidence that drives its outcome (the matched-surface assumption). The target of each scenario is performance averaged over incidence surfaces; for example,  $\text{MSE} = E_X E_{Z|X} E_\epsilon [(\hat{\tau} - \tau)^2]$ , with  $X$  drawn from the configuration's generator,  $Z$  from the design given  $X$ , and  $\epsilon \sim N(0, I)$ . Bias, coverage, and power are defined in the same way.

## Treatment assignment designs

We implemented eight treatment assignment designs and present six of them as the primary comparison set in this chapter. The other two, Saturation Quadrants and Balanced Halves, each share the mechanism of a retained design, and in the eight-design comparison at  $\tau = 1$  neither

could be distinguished from its counterpart under either neighbor type, by the Nemenyi and  
 Holm-adjusted Wilcoxon signed-rank tests described under Simulation design, resampling, and  
 metrics below, applied here to all eight designs. Saturation Quadrants assigns the saturations  
 20%, 40%, 60%, and 80% to the four quadrants in random order, without reference to incidence.  
 Its mean squared error was 0.0900, against 0.0907 for Incidence-Guided Saturation Quadrants,  
 under queen contiguity (Nemenyi  $p = 0.98$ ; Holm-adjusted Wilcoxon signed-rank  $p = 0.41$ ),  
 and 0.0708 against 0.0717 under rook contiguity (Nemenyi  $p = 1.00$ ; Wilcoxon  $p = 0.82$ ). We  
 retain one representative of the saturation mechanism, Incidence-Guided Saturation Quadrants.  
 Saturation Quadrants had the marginally lower mean under both neighbor types and the better  
 average rank in the eight-design comparison (2.100 against 2.350 under queen contiguity and  
 2.281 against 2.306 under rook contiguity), so this choice does not rest on performance; that  
 ordering the saturations by incidence makes no detectable difference is itself a result of this  
 chapter. Balanced Halves divides clusters into two halves by incidence rank and treats 25 at  
 random in each. Its mean squared error was 0.1299, against 0.1308 for Balanced Quartiles, under  
 queen contiguity (Nemenyi  $p = 1.00$ ; Wilcoxon  $p = 0.88$ ), and 0.0842 against 0.0837 under rook  
 contiguity (Nemenyi  $p = 1.00$ ; Wilcoxon  $p = 0.82$ ). The two designs differ only in how finely  
 they stratify on incidence, and we retain the finer, four-stratum version. Balanced Halves had  
 the marginally lower mean and the better average rank under queen contiguity (3.312 against  
 3.356), and Balanced Quartiles had both under rook contiguity (2.919 against 3.038), so this  
 choice does not rest on performance either. Both excluded designs, and the full eight-design  
 comparison, are reported in the appendix of this chapter.

Table 1 describes the six retained designs. Checkerboard is the strict block-stratified allocation  
 of Chapter 2, in which no two rook neighbors share an arm. Chapter 2's Block Stratified  
 Sampling design also admitted relaxed allocations on its  $3 \times 3$  grid, where the arms cannot be  
 balanced; on its  $2 \times 4$  and  $3 \times 4$  grids, as on the  $10 \times 10$  grid here, the strict allocations are the  
 two checkerboard patterns. We use the two names interchangeably for this allocation and use

Checkerboard in the rest of this chapter. Checkerboard is the only deterministic design: it gives the same assignment in every draw, although each of its fits still receives its own outcome noise. Every other design was drawn independently 25 times for each incidence surface. Designs that rank clusters by incidence break ties at random, afresh for each draw. High Incidence Focus is random only through that tie-breaking; under the iid and spatial mechanisms incidence values are distinct, so it gives the same assignment in all 25 draws on a surface, and its assignment changes only from one surface to the next. Isolation Buffer builds its assignment from the neighbor type of the scenario, and it treats 38 clusters on average under rook contiguity and 21 under queen contiguity. Every other design treats exactly 50.

Table 1: The six treatment assignment designs compared in this chapter, ordered by mean squared error under queen contiguity at  $\tau = 1$ , pooled over the five incidence configurations (Table 2). A cluster's incidence rank is its rank by the baseline incidence surface  $X_k$ , with ties broken at random in every draw. Clusters treated is the number of clusters treated per assignment; for Isolation Buffer, which depends on the neighbor type, it is the mean over its assignments under each neighbor type.

| Design                                                                  | Assignment rule                                                                                                                                                                                                                                                   | Clusters treated         | Deterministic |
|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|---------------|
| Incidence-Guided Saturation Quadrants                                   | The grid is divided into its four $5 \times 5$ coordinate quadrants. Quadrants are ranked by mean incidence and given saturations of 80%, 60%, 40%, and 20% from highest to lowest mean; within each quadrant, that share of its 25 clusters is treated at random | 50                       | No            |
| Balanced Quartiles                                                      | Clusters are divided by incidence rank into four strata of 25. Twelve clusters are treated at random in each stratum, and one more in each of two randomly chosen strata                                                                                          | 50                       | No            |
| Isolation Buffer                                                        | Clusters are treated one at a time, each chosen at random from those that are neither treated nor adjacent to a treated cluster, until none remain (a greedy random maximal independent set); no two treated clusters are neighbors                               | 38 rook, 21 queen (mean) | No            |
| High Incidence Focus                                                    | The 50 clusters with the highest incidence rank are treated                                                                                                                                                                                                       | 50                       | No            |
| 2×2 Blocking                                                            | The grid is divided into 25 non-overlapping 2×2 blocks; two of the four clusters in each block are treated at random                                                                                                                                              | 50                       | No            |
| Checkerboard (strict Block Stratified Sampling allocation in Chapter 2) | Clusters whose grid coordinates $(x, y)$ have an odd sum $x + y$ are treated, so treatment alternates from cluster to cluster; under rook contiguity every neighbor of a treated cluster is a control, and every neighbor of a control is treated                 | 50                       | Yes           |

## 203 Estimation

204 We estimated  $\tau$  with two maximum likelihood spatial lag models, both fit to every simulated  
205 outcome and both including an intercept. The primary, oracle model has the spatial lag  $\rho WY$   
206 and the covariates  $Z$ ,  $\text{Spill} = S(Z)$ , and  $X_k$ , so it matches the outcome model in Equation 1,  
207 including the true regime-specific spillover covariate from Equation 2. The oracle model assumes  
208 that the analyst knows the form of the spillover mechanism and has the exact exposure term, so  
209 its results are a best case for each design; we return to this assumption as a limitation. The  
210 non-oracle model, a sensitivity analysis, has the covariates  $Z$  and  $X_k$  only and omits the spillover  
211 term. Both models are fit to the same outcomes.

212 We fit both models with our own implementation of the estimator computed by the `lagsarlm`  
213 function of the R package `spatialreg` with `method = "eigen"` [25]. It maximizes the concen-  
214 trated log-likelihood over  $\rho$ , computing the log-determinant from the eigenvalues of  $W$  [12],  
215 estimates the remaining coefficients by least squares given  $\hat{\rho}$ , and computes standard errors from  
216 the analytic information matrix. On 5,120 validation fits it agreed with `lagsarlm` to within  
217  $9.7 \times 10^{-8}$  in  $\hat{\tau}$  and to a relative difference of  $1.1 \times 10^{-8}$  in its standard error, at roughly 1% of  
218 the computing time; we also refit a 1% sample of the full simulation with `lagsarlm` itself as a  
219 check. Confidence intervals are 95% Wald intervals,  $\hat{\tau} \pm 1.96 \widehat{\text{SE}}$ .

220 Two combinations of design and neighbor type weaken or remove the information that separates  
221  $\tau$  from the spillover term. Under rook contiguity, Checkerboard gives  $WZ = 1 - Z$  exactly,  
222 because every neighbor of a treated cluster is a control and every neighbor of a control is treated.  
223 Both spillover covariates then equal  $\gamma(1 - Z)$ , a linear combination of the intercept and  $Z$ , so  $\tau$   
224 is not identified: the fitting routine drops the aliased spillover column, and the coefficient on  $Z$   
225 estimates  $\tau - \gamma$ . Chapter 2 noted the same consequence of exact collinearity between treatment  
226 and spillover exposure,  $\hat{\beta} \approx \beta - \psi$  in its notation. The non-oracle model has no spillover column  
227 to drop, but its Checkerboard estimates under rook contiguity have the same defect. Under

228 queen contiguity, each of the 64 interior clusters has four treated and four control neighbors,  
 229 so  $WZ = 1/2$  at every interior cluster. Under the control-only regime the spillover covariate  
 230 then equals  $(\gamma/2)(1 - Z)$  at those clusters, and only the 36 clusters on the grid boundary carry  
 231 information that separates it from the intercept and  $Z$ ;  $\tau$  is identified, but imprecisely. Under the  
 232 both-arms regime it is the constant  $\gamma/2$  at interior clusters, confounded with the intercept but  
 233 not with  $Z$ . Isolation Buffer under rook contiguity can also, rarely, draw exactly the checkerboard  
 234 pattern, a maximal independent set that equals one color class of the grid. This happened  
 235 in 4 of its 40,000 rook draws and affected one fit in each of 20 scenarios. We flag every fit in  
 236 which the fitting routine reports an aliased column and every scenario in which some draw has  
 237  $\text{rank}([1, Z, WZ]) < 3$ . Aliased estimates are kept in all summaries and flagged in exhibits.

## 238 **Simulation design, resampling, and metrics**

239 The full simulation crossed five values of  $\tau$ , five incidence configurations, two neighbor types, four  
 240 values of  $\rho$ , four values of  $\gamma$ , two spillover regimes, and eight designs, for  $5 \times 5 \times 2 \times 4 \times 4 \times 2 \times 8 =$   
 241 12,800 scenarios, of which the six designs reported in this chapter account for 9,600. We call a  
 242 combination of incidence configuration, neighbor type,  $\rho$ ,  $\gamma$ , and spillover regime a block; there  
 243 are  $5 \times 2 \times 4 \times 4 \times 2 = 320$  blocks, 160 for each neighbor type. At the primary scenario,  $\tau = 1$ ,  
 244 the six designs give 1,920 scenarios, and every block contains all six.

245 Within each scenario, we drew  $J = 25$  assignments from each of the  $K = 10$  incidence surfaces,  
 246 for 250 fits per scenario, and gave every fit its own noise vector, so that fit  $(k, j)$  has outcome  
 247  $Y_{kj} = (I - \rho W)^{-1}[\tau Z_{kj} + S(Z_{kj}) + \beta X_k + \epsilon_{kj}]$ . Checkerboard's 250 fits share one assignment  
 248 but have 250 distinct noise vectors. Given a surface, its 25 pairs of assignment and noise are  
 249 independent, and the ten surfaces are independent of one another, so the surface is the primary  
 250 sampling unit. Each estimator was fit 3.2 million times.

251 Every random draw was seeded from a hash of a key that names the draw: the configuration and  
 252 surface index for an incidence surface; the block, surface, and design for a set of 25 assignments;

and the block for the noise. Each draw therefore depends only on its key, and the results are identical whether scenarios are run sequentially, in parallel, or in any order. The keys also give common random numbers. Within a block, all designs share the same noise vectors and incidence surfaces; the five values of  $\tau$  reuse the same assignments and noise; and the oracle and non-oracle models are fit to the same outcomes. Assignments and noise are independent across blocks, but blocks within an incidence configuration share its ten surfaces.

For each scenario and estimator, we computed five metrics over the valid fits, those with a finite estimate and standard error: bias, the mean of  $\hat{\tau} - \tau$ ; the standard deviation (SD) of  $\hat{\tau}$ ; mean squared error,  $\text{MSE} = \text{mean}\{(\hat{\tau} - \tau)^2\}$ ; coverage, the fraction of fits with  $L \leq \tau \leq U$ , where  $L, U = \hat{\tau} \mp 1.96 \widehat{\text{SE}}$ ; and power, the fraction of fits with  $L > 0$ , a one-sided test of  $\tau = 0$  at level 0.025. For both estimators, every one of the 250 fits in every scenario was valid, and the fitting routine raised no warning other than the aliasing flags described above.

Monte Carlo standard errors come from the surface-level metrics. For a metric  $m$ , we computed  $m_k$  on the 25 fits from surface  $k$  and set  $\text{SE}(m) = \text{sd}(m_1, \dots, m_{10})/\sqrt{10}$ , with intervals  $m \pm t_{0.975, 9} \text{SE}(m)$ . A difference between two designs is paired at the surface level: its standard error is that of  $m_{d,k} - m_{d',k}$  over  $k$ . For a metric pooled over the five incidence configurations, we apply the same formula to the 50 configuration-by-surface means, each averaged over that surface's 32 blocks; these 50 means are independent of one another.

We compared designs within each neighbor type with a Friedman rank-sum test on scenario MSE [26], treating the 160 blocks as repeated measures across the six designs. At  $\tau = 1$  we followed it with Nemenyi post-hoc comparisons [27, 28] and paired Wilcoxon signed-rank tests [29] with Holm's adjustment for multiple comparisons [30] for all 15 pairs of designs, and we repeated the Friedman test at each other value of  $\tau$ . Blocks within an incidence configuration share their surfaces, so these tests treat dependent blocks as independent, which overstates the information about variation from surface to surface. For each pair of designs adjacent in the

ranking, we therefore also ran a paired Wilcoxon signed-rank test over 50 independent units, the 5 configurations times 10 surfaces, with each unit's MSE averaged over that surface's blocks, and a  $t_{49}$  interval for the mean difference. Because the values of  $\tau$  share assignments and noise, Friedman tests at different values of  $\tau$  are not independent confirmations of one another.

We report results separately for each neighbor type, with queen contiguity primary, and separately for each incidence configuration. Any number averaged over incidence configurations or over neighbor types is labeled as pooled, and we show the split wherever a conclusion changes.

## Results

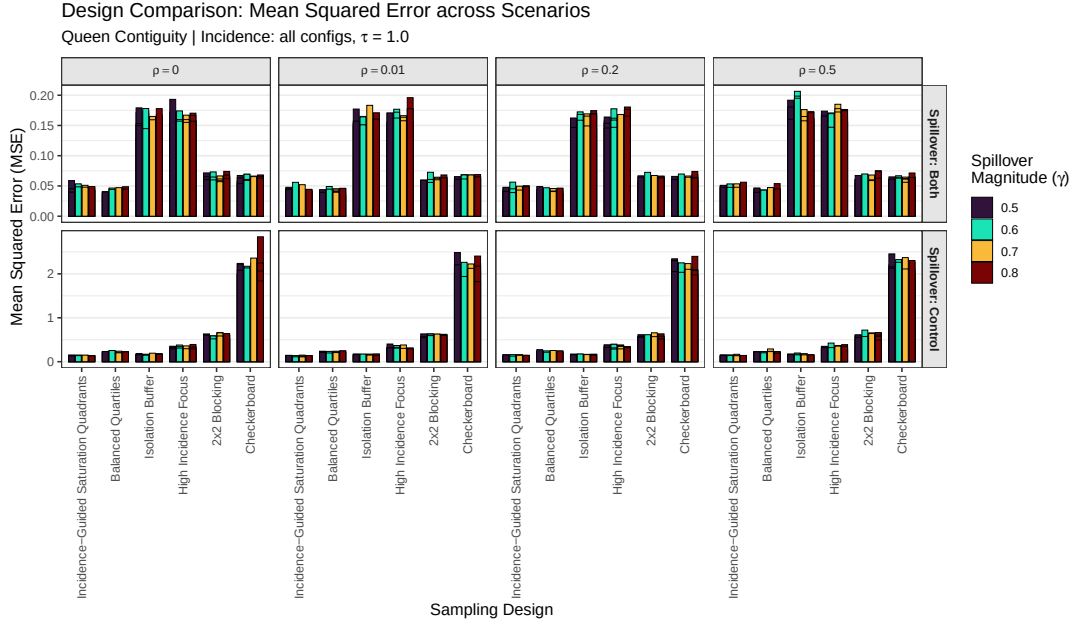
### Design performance under queen contiguity

At the primary scenario,  $\tau = 1$ , under queen contiguity and pooled over the five incidence configurations, Incidence-Guided Saturation Quadrants had the lowest mean squared error (MSE = 0.091), followed by Balanced Quartiles (0.131), Isolation Buffer (0.160), High Incidence Focus (0.245),  $2 \times 2$  Blocking (0.319), and Checkerboard (1.087) (Table 2, Figure 1a). Coverage of the 95% intervals was between 0.936 and 0.941 for all six designs (Figure 2a), so under queen contiguity the designs differed in precision, not in the validity of their intervals. Power to reject  $\tau = 0$  was 0.930 for Incidence-Guided Saturation Quadrants, 0.827 for Balanced Quartiles, 0.731 for Isolation Buffer, 0.642 for  $2 \times 2$  Blocking, 0.610 for High Incidence Focus, and 0.547 for Checkerboard.

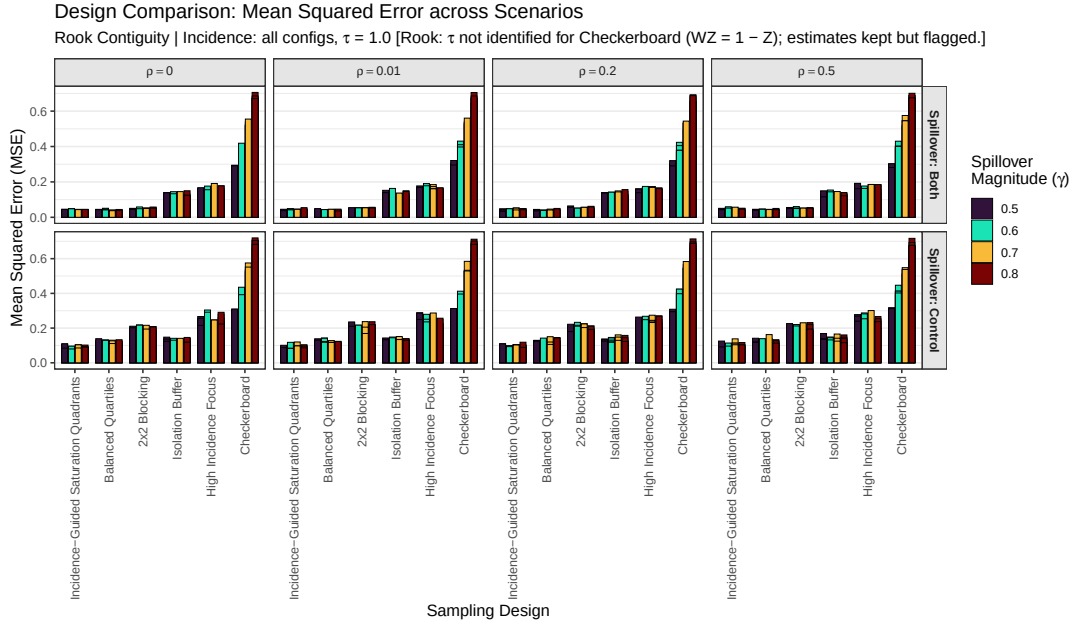
Incidence-Guided Saturation Quadrants also led block by block. Of the 160 queen blocks at  $\tau = 1$ , Incidence-Guided Saturation Quadrants had the lowest MSE in 95 (59%), Balanced Quartiles in 58 (36%), and Isolation Buffer in 7 (4%); no other design had the lowest MSE in any block.

Table 2: Performance of the six designs at the primary scenario,  $\tau = 1$ , under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages the 160 scenarios of one design and neighbor type (5 incidence configurations  $\times$  4 values of  $\rho \times$  4 values of  $\gamma \times$  2 spillover regimes), each summarizing 250 fits of the oracle model. Rows in both panels are in order of mean squared error (MSE) under queen contiguity. Monte Carlo standard errors of MSE, in parentheses, are computed from the 50 configuration-by-surface means (Methods); those of bias, coverage, and power are all below 0.007. Average rank is the mean of each design’s within-block MSE rank among the six designs over the 160 blocks (1 = lowest MSE). Clusters treated is the mean number of clusters treated per assignment. Under rook contiguity,  $\tau$  is not identified for Checkerboard ( $WZ = 1 - Z$ ); its estimates are kept and flagged, and its coefficient on  $Z$  estimates  $\tau - \gamma$ .

| Design                               | MSE (SE)       | Bias   | Coverage | Power | Average rank | Clusters treated |
|--------------------------------------|----------------|--------|----------|-------|--------------|------------------|
| <i>Queen contiguity (primary)</i>    |                |        |          |       |              |                  |
| Incidence-Guided                     | 0.091 (0.0008) | 0.032  | 0.937    | 0.930 | 1.425        | 50               |
| Saturation Quadrants                 |                |        |          |       |              |                  |
| Balanced Quartiles                   | 0.131 (0.0011) | 0.017  | 0.936    | 0.827 | 2.138        | 50               |
| Isolation Buffer                     | 0.160 (0.0011) | 0.003  | 0.939    | 0.731 | 3.694        | 21.4             |
| High Incidence Focus                 | 0.245 (0.0047) | 0.031  | 0.938    | 0.610 | 4.763        | 50               |
| 2 $\times$ 2 Blocking                | 0.319 (0.0032) | 0.010  | 0.939    | 0.642 | 4.244        | 50               |
| Checkerboard                         | 1.087 (0.0101) | 0.001  | 0.941    | 0.547 | 4.737        | 50               |
| <i>Rook contiguity (sensitivity)</i> |                |        |          |       |              |                  |
| Incidence-Guided                     | 0.072 (0.0006) | 0.018  | 0.942    | 0.963 | 1.488        | 50               |
| Saturation Quadrants                 |                |        |          |       |              |                  |
| Balanced Quartiles                   | 0.084 (0.0006) | 0.008  | 0.940    | 0.917 | 1.775        | 50               |
| Isolation Buffer                     | 0.133 (0.0009) | 0.003  | 0.939    | 0.806 | 3.381        | 38.2             |
| High Incidence Focus                 | 0.206 (0.0030) | 0.018  | 0.939    | 0.645 | 4.969        | 50               |
| 2 $\times$ 2 Blocking                | 0.128 (0.0012) | 0.004  | 0.938    | 0.818 | 3.400        | 50               |
| Checkerboard (flagged)               | 0.480 (0.0015) | −0.652 | 0.150    | 0.414 | 5.987        | 50               |

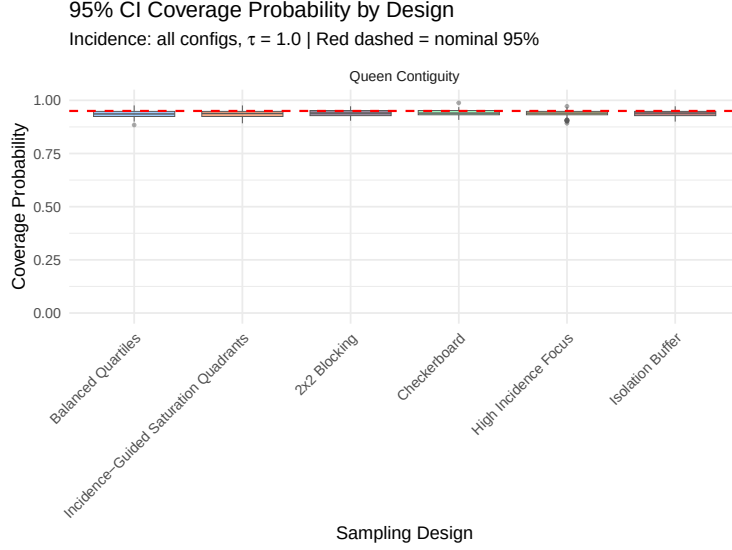


(a) Queen contiguity (primary)

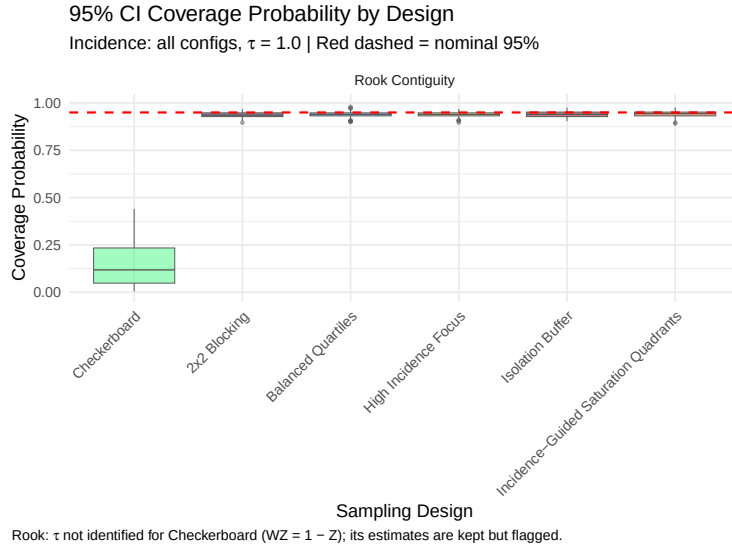


(b) Rook contiguity (sensitivity)

Figure 1: Mean squared error (MSE) of the oracle estimate of  $\tau$  by design at  $\tau = 1$ , under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Columns of panels correspond to the four levels of outcome spatial dependence  $\rho$ , rows to the both-arms (top) and control-only (bottom) spillover regimes, and bar colors to the spillover magnitude  $\gamma$ . Each bar overlays the MSEs of the five incidence configurations for one combination of design,  $\rho$ ,  $\gamma$ , and regime: the bar's height is the largest of the five, and the horizontal lines within it mark the other four. Designs are ordered by pooled mean MSE for the neighbor type of the panel (Table 2). The vertical scale is set separately for each row. Under rook contiguity,  $\tau$  is not identified for Checkerboard; its estimates are kept and flagged.



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure 2: Coverage of the 95% confidence intervals by design at  $\tau = 1$ , under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Each box summarizes the coverage of the 160 scenarios of one design (5 incidence configurations  $\times$  4 values of  $\rho \times$  4 values of  $\gamma \times$  2 spillover regimes), each estimated from 250 fits. Designs are ordered by median coverage within each panel, so the order differs between panels and from the MSE order used elsewhere. The dashed red line marks the nominal 0.95 level. Under rook contiguity,  $\tau$  is not identified for Checkerboard; its estimates are kept and flagged.

## Formal statistical comparison

The Friedman test rejected equal MSE ranks across the six designs under queen contiguity at  $\tau = 1$  ( $\chi^2 = 451.57$ ,  $\text{df} = 5$ , 160 blocks,  $p < 2.2 \times 10^{-16}$ ). The average within-block ranks were 1.43 for Incidence-Guided Saturation Quadrants, 2.14 for Balanced Quartiles, 3.69 for Isolation Buffer, 4.24 for  $2 \times 2$  Blocking, 4.74 for Checkerboard, and 4.76 for High Incidence Focus. Nemenyi comparisons (critical difference 0.596) separated 11 of the 15 pairs. Incidence-Guided Saturation Quadrants ranked significantly better than every other design, including Balanced Quartiles ( $p = 0.0086$ ), and Balanced Quartiles ranked significantly better than the remaining four. The four pairs that Nemenyi comparisons did not separate all lay among the bottom four designs:  $2 \times 2$  Blocking against Isolation Buffer ( $p = 0.090$ ), High Incidence Focus ( $p = 0.130$ ), and Checkerboard ( $p = 0.170$ ), and Checkerboard against High Incidence Focus ( $p = 1.00$ ). Holm-adjusted Wilcoxon signed-rank tests separated all 15 pairs.

The bottom three designs were ordered differently by average rank and by mean MSE. By mean MSE the order was High Incidence Focus (0.245),  $2 \times 2$  Blocking (0.319), and Checkerboard (1.087); by average rank it was  $2 \times 2$  Blocking (4.24), Checkerboard (4.74), and High Incidence Focus (4.76). Both reversals have the same source: the MSEs of  $2 \times 2$  Blocking and Checkerboard depended strongly on the spillover regime. Each had a lower MSE than High Incidence Focus in all 80 both-arms blocks (mean MSE 0.062 for each, against 0.162) and a higher MSE in all 80 control-only blocks (0.576 and 2.111, against 0.329; see the spillover-regime results below). Average rank counts each block once, so their wins under the both-arms regime offset their losses under the control-only regime; mean MSE weights those losses by their size.

Because blocks within an incidence configuration share their ten surfaces, we also compared each pair of designs adjacent in the mean-MSE order over the 50 independent configuration-by-surface units. Every adjacent pair differed significantly. The mean paired differences in MSE, with  $t_{49}$  95% intervals, were  $-0.0401$  ( $-0.0431$ ,  $-0.0372$ ) for Incidence-Guided Saturation Quadrants

against Balanced Quartiles,  $-0.0293$  ( $-0.0332$ ,  $-0.0255$ ) for Balanced Quartiles against Isolation Buffer,  $-0.0853$  ( $-0.0954$ ,  $-0.0752$ ) for Isolation Buffer against High Incidence Focus,  $-0.0738$  ( $-0.0858$ ,  $-0.0617$ ) for High Incidence Focus against  $2 \times 2$  Blocking, and  $-0.7677$  ( $-0.7893$ ,  $-0.7460$ ) for  $2 \times 2$  Blocking against Checkerboard, with Wilcoxon  $p \leq 4.9 \times 10^{-9}$  in every case. The better design of each pair had the lower MSE in all 50 units, except High Incidence Focus against  $2 \times 2$  Blocking, where it did so in 48 (96%). This test follows the mean-MSE order. It separated High Incidence Focus from  $2 \times 2$  Blocking, which the Nemenyi comparison of within-block ranks did not, because each unit's MSE averages over both spillover regimes, and there the control-only losses of  $2 \times 2$  Blocking outweigh its both-arms wins.

### Design rankings by incidence configuration

Within each of the five incidence configurations, the queen mean-MSE order was the same as the pooled order: Incidence-Guided Saturation Quadrants, Balanced Quartiles, Isolation Buffer, High Incidence Focus,  $2 \times 2$  Blocking, and Checkerboard. Across the five configurations, the MSE of Incidence-Guided Saturation Quadrants ranged from 0.087 (spatial,  $\rho_X = 0.20$ ) to 0.093 (iid), that of Balanced Quartiles from 0.128 to 0.135, Isolation Buffer from 0.159 to 0.162, High Incidence Focus from 0.232 to 0.267,  $2 \times 2$  Blocking from 0.311 to 0.326, and Checkerboard from 1.068 to 1.112. Incidence-Guided Saturation Quadrants had the lowest MSE in 18 to 21 of the 32 blocks in each configuration. Coverage did not depend on the configuration either: within each configuration, every design's coverage, averaged over its 32 scenarios, was between 0.931 and 0.943. The mechanism that generated the incidence surface therefore changed each design's MSE only slightly and did not change the ranking. This includes High Incidence Focus, the design that uses the incidence ranks most directly: its MSE under the two Poisson configurations (0.232 and 0.238) was no higher than under the iid (0.245) and spatial (0.267 and 0.245) configurations.

## Sensitivity to spatial dependence, spillover magnitude, and spillover regime

Under queen contiguity, neither the strength of spatial dependence in the outcome nor the spillover magnitude changed the ranking. Across  $\rho \in \{0, 0.01, 0.20, 0.50\}$ , no design's MSE changed by more than 0.02 (Table 3). Across  $\gamma \in \{0.5, 0.6, 0.7, 0.8\}$ , the MSE of every design except Checkerboard also varied by less than 0.02, and Checkerboard's varied between 1.052 and 1.108 with no trend in  $\gamma$  (Table 4). Under rook contiguity,  $\gamma$  mattered for Checkerboard alone, for the reason given in the rook results below.

Table 3: Mean squared error by design across the four levels of outcome spatial dependence  $\rho$ , under queen contiguity at  $\tau = 1$ , pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations  $\times$  4 values of  $\gamma \times 2$  spillover regimes). Rows are ordered by pooled mean MSE under queen contiguity (Table 2).

| Design                                | $\rho = 0$ | $\rho = 0.01$ | $\rho = 0.20$ | $\rho = 0.50$ |
|---------------------------------------|------------|---------------|---------------|---------------|
| Incidence-Guided Saturation Quadrants | 0.090      | 0.087         | 0.093         | 0.093         |
| Balanced Quartiles                    | 0.129      | 0.129         | 0.132         | 0.133         |
| Isolation Buffer                      | 0.161      | 0.157         | 0.158         | 0.165         |
| High Incidence Focus                  | 0.243      | 0.242         | 0.249         | 0.248         |
| 2 $\times$ 2 Blocking                 | 0.318      | 0.320         | 0.314         | 0.324         |
| Checkerboard                          | 1.095      | 1.077         | 1.084         | 1.092         |

Table 4: Mean squared error by design across the four levels of spillover magnitude  $\gamma$  at  $\tau = 1$ , under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations  $\times$  4 values of  $\rho \times 2$  spillover regimes). Rows within each panel are ordered by that neighbor type's pooled mean MSE (Table 2). Under rook contiguity,  $\tau$  is not identified for Checkerboard; its estimates are kept and flagged.

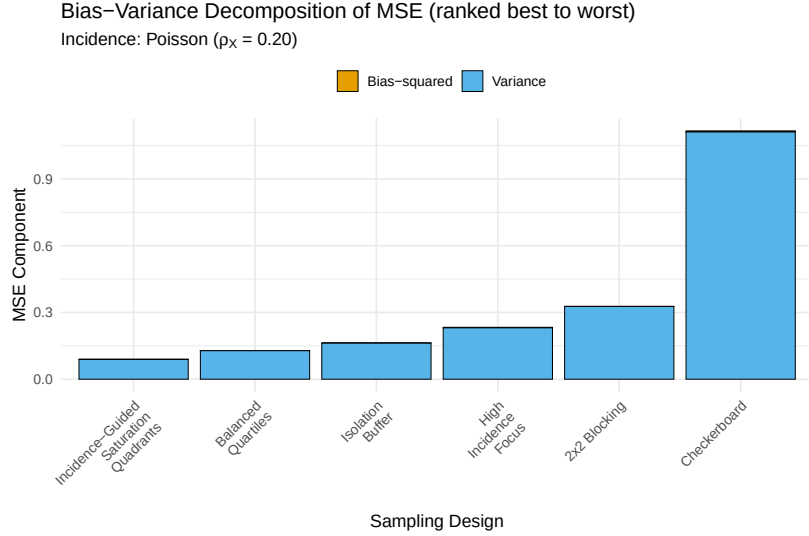
| Design                                | $\gamma = 0.5$ | $\gamma = 0.6$ | $\gamma = 0.7$ | $\gamma = 0.8$ |
|---------------------------------------|----------------|----------------|----------------|----------------|
| <i>Queen contiguity</i>               |                |                |                |                |
| Incidence-Guided Saturation Quadrants | 0.092          | 0.089          | 0.092          | 0.089          |
| Balanced Quartiles                    | 0.132          | 0.128          | 0.132          | 0.131          |
| Isolation Buffer                      | 0.159          | 0.163          | 0.160          | 0.159          |
| High Incidence Focus                  | 0.244          | 0.249          | 0.248          | 0.241          |
| 2 $\times$ 2 Blocking                 | 0.312          | 0.314          | 0.328          | 0.323          |
| Checkerboard                          | 1.108          | 1.052          | 1.086          | 1.101          |
| <i>Rook contiguity</i>                |                |                |                |                |
| Incidence-Guided Saturation Quadrants | 0.069          | 0.072          | 0.075          | 0.072          |
| Balanced Quartiles                    | 0.083          | 0.085          | 0.083          | 0.083          |
| 2 $\times$ 2 Blocking                 | 0.129          | 0.129          | 0.127          | 0.128          |
| Isolation Buffer                      | 0.131          | 0.133          | 0.134          | 0.133          |
| High Incidence Focus                  | 0.207          | 0.209          | 0.205          | 0.203          |
| Checkerboard (flagged)                | 0.295          | 0.403          | 0.538          | 0.684          |

The spillover regime was the one factor that changed the queen ranking (Table 5, Figure 1a).

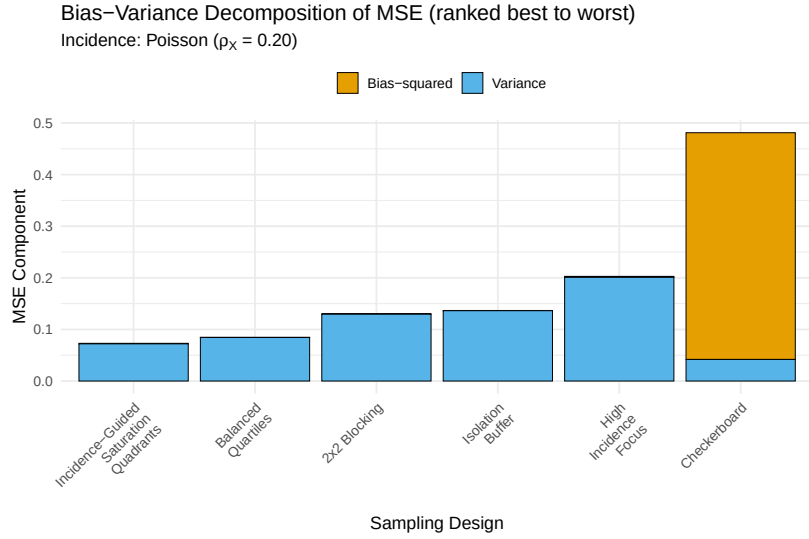
Under the both-arms regime, Balanced Quartiles had the lowest MSE (0.042), followed by Incidence-Guided Saturation Quadrants (0.046),  $2 \times 2$  Blocking (0.062), Checkerboard (0.062), Isolation Buffer (0.161), and High Incidence Focus (0.162). Under the control-only regime, the order was Incidence-Guided Saturation Quadrants (0.135), Isolation Buffer (0.159), Balanced Quartiles (0.219), High Incidence Focus (0.329),  $2 \times 2$  Blocking (0.576), and Checkerboard (2.111). Isolation Buffer’s MSE was the same under both regimes (0.161 and 0.159) by construction: no treated cluster has a treated neighbor, so  $WZ = 0$  at every treated cluster,  $WZ \circ (1 - Z) = WZ$ , and the two regimes in Equation 2 define the same outcome model; the small difference is Monte Carlo error, since the two regimes use separate draws. Every other design had a higher MSE under the control-only regime.

That increase was variance, not bias. The oracle model includes the true regime-specific spillover covariate, so it has no omitted-variable bias. In every combination of design and regime under queen contiguity, the absolute bias was at most 0.048 and squared bias, averaged over scenarios, was at most 2.2% of MSE. Checkerboard showed the mechanism most clearly. At its 64 interior clusters,  $WZ = 1/2$ . Under the both-arms regime the spillover covariate is therefore the constant  $\gamma/2$  at those clusters, confounded with the intercept but not with  $Z$ . Under the control-only regime it equals  $(\gamma/2)(1 - Z)$  there, a linear combination of the intercept and  $Z$ , so only the 36 boundary clusters separate the spillover effect from the treatment effect. Checkerboard’s MSE rose from 0.062 under the both-arms regime to 2.111 under the control-only regime, while its bias stayed near zero (0.004) and its coverage stayed at 0.942: its intervals widened with its variance. Its pooled MSE of 1.087 was almost entirely variance: squared bias, averaged over scenarios, was 0.0045, less than 0.5% of MSE (Figure 3a shows the decomposition for the Poisson configuration with  $\rho_X = 0.20$ ).  $2 \times 2$  Blocking showed the same pattern on a smaller scale, with MSE rising from 0.062 to 0.576 and a bias of 0.016 under the control-only regime.

Two designs were therefore fragile with respect to the spillover regime, which a trial team cannot



(a) Queen contiguity (primary)



(b) Rook contiguity (sensitivity)

Figure 3: Decomposition of mean squared error into squared bias and variance by design at  $\tau = 1$  for one incidence configuration, Poisson with  $\rho_X = 0.20$ , under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). For each design, the two stacked components are the squared bias and the variance ( $SD^2$ ) of  $\hat{\tau}$ , each averaged over the 32 scenarios of that configuration (4 values of  $\rho \times 4$  values of  $\gamma \times 2$  spillover regimes). Designs are ordered by pooled mean MSE for the neighbor type of the panel (Table 2). The vertical scales of the two panels differ. Under rook contiguity,  $\tau$  is not identified for Checkerboard; its estimates are kept and flagged, and nearly all of its MSE is squared bias.

Table 5: Mean squared error by design under each spillover regime at  $\tau = 1$ , under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 80 scenarios (5 incidence configurations  $\times$  4 values of  $\rho \times$  4 values of  $\gamma$ ). Rows are ordered by pooled mean MSE under queen contiguity (Table 2). Under rook contiguity,  $\tau$  is not identified for Checkerboard; its estimates are kept and flagged.

| Design                                | Queen     |              | Rook      |              |
|---------------------------------------|-----------|--------------|-----------|--------------|
|                                       | Both arms | Control only | Both arms | Control only |
| Incidence-Guided Saturation Quadrants | 0.046     | 0.135        | 0.045     | 0.098        |
| Balanced Quartiles                    | 0.042     | 0.219        | 0.042     | 0.125        |
| Isolation Buffer                      | 0.161     | 0.159        | 0.131     | 0.134        |
| High Incidence Focus                  | 0.162     | 0.329        | 0.164     | 0.249        |
| 2 $\times$ 2 Blocking                 | 0.062     | 0.576        | 0.051     | 0.206        |
| Checkerboard (flagged under rook)     | 0.062     | 2.111        | 0.477     | 0.483        |

know in advance. Under queen contiguity, Checkerboard and  $2 \times 2$  Blocking were close to the best designs under the both-arms regime and were the two worst under the control-only regime. Incidence-Guided Saturation Quadrants was first or second under both regimes.

### Rook contiguity

Under rook contiguity at  $\tau = 1$ , pooled over the five incidence configurations, the mean-MSE order was Incidence-Guided Saturation Quadrants (0.072), Balanced Quartiles (0.084),  $2 \times 2$  Blocking (0.128), Isolation Buffer (0.133), High Incidence Focus (0.206), and Checkerboard (0.480) (Table 2, Figure 1b), and the same order held within each of the five incidence configurations. Incidence-Guided Saturation Quadrants had the lowest MSE in 97 of the 160 rook blocks (61%) and Balanced Quartiles in 60 (38%). Coverage was between 0.938 and 0.942 for every design except Checkerboard, whose coverage was 0.150 (Figure 2b); within single incidence configurations, the corresponding ranges were 0.935 to 0.946, and 0.14 to 0.15 for Checkerboard. The Friedman test rejected equal ranks ( $\chi^2 = 703.76$ ,  $df = 5$ , 160 blocks,  $p < 2.2 \times 10^{-16}$ ). Nemenyi comparisons separated 13 of 15 pairs; the two they did not separate were Incidence-Guided Saturation Quadrants against Balanced Quartiles ( $p = 0.74$ ) and  $2 \times 2$  Blocking against Isolation Buffer ( $p = 1.00$ ). Holm-adjusted Wilcoxon tests separated every pair except  $2 \times 2$  Blocking against Isolation Buffer. In the surface-level comparison, Incidence-Guided Saturation

398 Quadrants had a lower MSE than Balanced Quartiles in 48 of the 50 units (mean difference  
399  $-0.0121$ ;  $-0.0139$ ,  $-0.0102$ ; Wilcoxon  $p = 9.9 \times 10^{-10}$ ).  $2 \times 2$  Blocking and Isolation Buffer were  
400 close:  $2 \times 2$  Blocking had the lower MSE in 36 of the 50 units (72%; mean difference  $-0.0048$ ;  
401  $-0.0078$ ,  $-0.0017$ ; Wilcoxon  $p = 0.0027$ ), but Isolation Buffer had the slightly better average  
402 rank (3.38 against 3.40), and the two designs traded places at some values of  $\tau$  (see the results  
403 for  $\tau$  below). Every other adjacent pair differed in all 50 units.

404 Checkerboard under rook contiguity is a different case:  $\tau$  is not identified, and the coefficient on  
405  $Z$  estimates  $\tau - \gamma$  (Methods). The estimates behaved accordingly. At  $\tau = 1$ , Checkerboard's  
406 mean bias was  $-0.502$ ,  $-0.600$ ,  $-0.704$ , and  $-0.802$  at  $\gamma = 0.5$ ,  $0.6$ ,  $0.7$ , and  $0.8$ , while its  
407 mean standard deviation stayed between  $0.20$  and  $0.21$ ; squared bias, averaged over scenarios,  
408 accounted for  $0.438$  of its MSE of  $0.480$  (Figure 3b shows the decomposition for the Poisson  
409 configuration with  $\rho_X = 0.20$ ). Its coverage fell from  $0.316$  at  $\gamma = 0.5$  to  $0.027$  at  $\gamma = 0.8$ . Its  
410 MSE was the same under both regimes ( $0.477$  and  $0.483$ ), as expected when the two spillover  
411 covariates coincide. Under queen contiguity, by contrast, Checkerboard's coverage stayed between  
412  $0.939$  and  $0.943$  at every level of  $\rho$ ; under rook contiguity it was between  $0.134$  and  $0.161$  (Table  
413 6). Rook Checkerboard estimates were precise but centered on the wrong quantity, the opposite  
414 of its imprecise but valid queen estimates. We keep these estimates in every ranking and flag  
415 them.

Table 6: Checkerboard's 95% confidence interval coverage across the four levels of outcome spatial dependence  $\rho$  at  $\tau = 1$ , under queen contiguity (primary) and rook contiguity (sensitivity), pooled over the five incidence configurations. Each entry averages 40 scenarios (5 incidence configurations  $\times$  4 values of  $\gamma$   $\times$  2 spillover regimes). Under rook contiguity,  $\tau$  is not identified for Checkerboard; its estimates are kept and flagged. The nominal level is  $0.95$ .

| Neighbor type | $\rho = 0$ | $\rho = 0.01$ | $\rho = 0.20$ | $\rho = 0.50$ |
|---------------|------------|---------------|---------------|---------------|
| Queen         | 0.941      | 0.943         | 0.940         | 0.939         |
| Rook          | 0.161      | 0.156         | 0.148         | 0.134         |

## Non-oracle sensitivity analysis

The non-oracle model, which omits the spillover term, was biased for every design. At  $\tau = 1$ , and setting aside Checkerboard under rook contiguity, its bias ranged from  $-0.12$  to  $-0.40$  and its coverage from  $0.51$  to  $0.90$  across designs and neighbor types. Under queen contiguity its bias ranged from  $-0.123$  (Incidence-Guided Saturation Quadrants) to  $-0.244$  ( $2 \times 2$  Blocking) and its coverage from  $0.709$  ( $2 \times 2$  Blocking) to  $0.896$  (High Incidence Focus); under rook contiguity the most biased design was Isolation Buffer (bias  $-0.400$ , coverage  $0.513$ ). For Checkerboard under rook contiguity, the non-oracle estimates were identical to the oracle estimates (MSE  $0.480$ , bias  $-0.652$ , coverage  $0.150$ ), because once the aliased spillover column is dropped the two models are the same.

Omitting the spillover term also removes a column that can be nearly collinear with  $Z$ , and the non-oracle estimates were accordingly less variable for some designs. Under queen contiguity, the non-oracle MSE was lower than the oracle MSE for every design, most of all for Checkerboard (oracle  $1.087$ , non-oracle  $0.108$ ) and  $2 \times 2$  Blocking (oracle  $0.319$ , non-oracle  $0.121$ ). Under rook contiguity it was higher for every design except  $2 \times 2$  Blocking ( $0.128$  under both models) and Checkerboard (identical). These results do not change the recommendation. Incidence-Guided Saturation Quadrants also had the lowest non-oracle MSE under both neighbor types ( $0.087$  under queen contiguity and  $0.086$  under rook contiguity), and where a design's non-oracle MSE was lower, the estimate was biased and its intervals under-covered, so the lower MSE does not support valid inference.

## Stability across treatment effect sizes

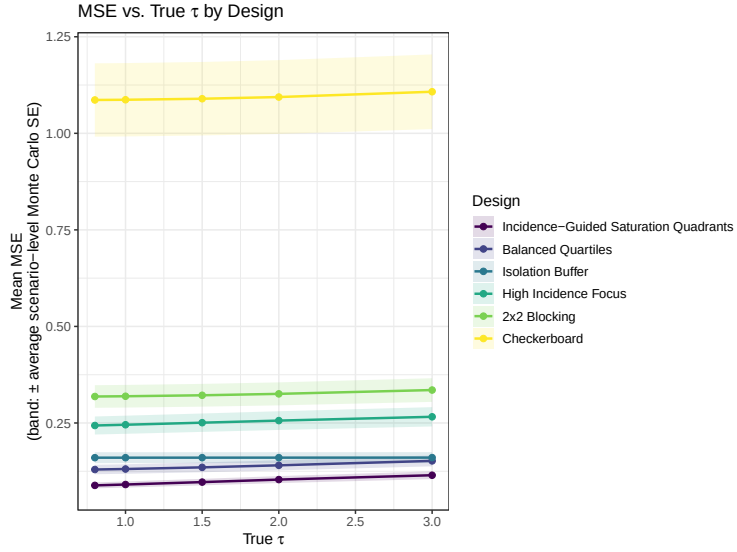
Under queen contiguity, each of the two design orders was the same at every  $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ : the order by average within-block rank did not change across these values, and neither did the order by mean MSE, although the two orders differ from each other in the bottom three designs (see Formal statistical comparison; Figure 4a shows mean

MSE). The Friedman statistic was 455.49, 451.57, 429.01, 411.19, and 376.79 at these five values ( $p < 2.2 \times 10^{-16}$  at each). By average rank, the lead of Incidence-Guided Saturation Quadrants over Balanced Quartiles narrowed as  $\tau$  increased: its average rank rose from 1.41 at  $\tau = 0.8$  to 1.85 at  $\tau = 3$ , while that of Balanced Quartiles stayed between 2.13 and 2.14. By mean MSE, the gap between the two changed little, staying between 0.037 and 0.041, because the MSE of both rose (from 0.088 to 0.115 for Incidence-Guided Saturation Quadrants and from 0.130 to 0.152 for Balanced Quartiles), while that of Isolation Buffer stayed at 0.160. Under rook contiguity the order was also stable (Figure 4b) except for the close pair of  $2 \times 2$  Blocking and Isolation Buffer: by average rank,  $2 \times 2$  Blocking was ahead only at  $\tau = 0.8$ , and by mean MSE, Isolation Buffer was ahead only at  $\tau = 3$ . The five values of  $\tau$  share their assignments and noise, so these five comparisons are not independent confirmations of one another; they show that, under queen contiguity, the order did not change across the five values of  $\tau$  we tested, given the same draws.

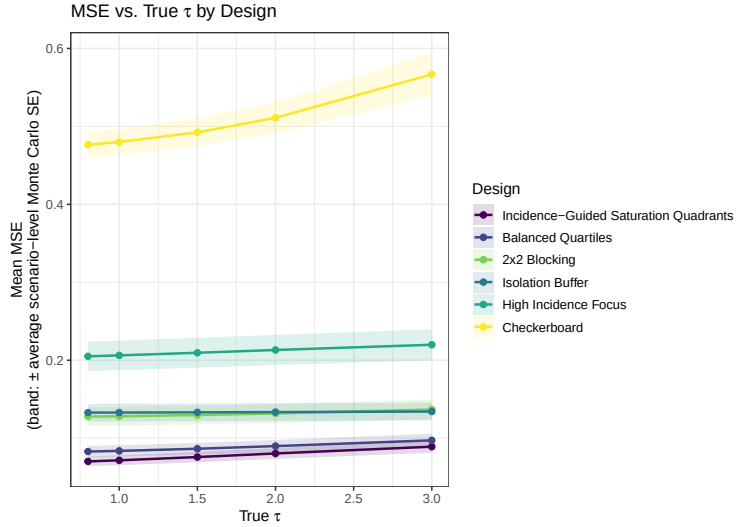
## Robustness across blocks and the recommended design

Table 7 gives the distribution of MSE across the 160 queen blocks at  $\tau = 1$ . Incidence-Guided Saturation Quadrants had the lowest median (0.080) and the lowest worst-case MSE of any design (0.169). Isolation Buffer had the narrowest range (0.128 to 0.206), because its MSE does not depend on the spillover regime, but at a higher level. Balanced Quartiles had a low median (0.115) and a worst case of 0.290. The distributions of  $2 \times 2$  Blocking and Checkerboard reflected their dependence on the spillover regime: both had low first quartiles (0.062 and 0.063) and high upper quartiles (0.580 and 2.093). Checkerboard had the highest MSE in all 80 control-only blocks; in the 80 both-arms blocks, the highest MSE belonged to High Incidence Focus in 42 and to Isolation Buffer in 38. Under rook contiguity, Incidence-Guided Saturation Quadrants also had the lowest worst-case MSE (0.138).

We recommend Incidence-Guided Saturation Quadrants. It had the lowest mean MSE under both



(a) Queen contiguity (primary)



Rook:  $\tau$  not identified for Checkerboard ( $WZ = 1 - Z$ ); its estimates are kept but flagged.

(b) Rook contiguity (sensitivity)

Figure 4: Mean squared error by design across the five values of the treatment effect,  $\tau \in \{0.8, 1.0, 1.5, 2.0, 3.0\}$ , under (a) queen contiguity (primary) and (b) rook contiguity (sensitivity). Each point averages the 160 scenarios of one design at one value of  $\tau$ , pooled over the five incidence configurations. Shaded bands show  $\pm 1$  times the average of the scenario-level Monte Carlo standard errors of MSE; they describe the precision of a single scenario's MSE, not of the plotted average, whose standard error is smaller (Table 2). The five values of  $\tau$  share their assignments and noise. The vertical scales of the two panels differ. Under rook contiguity,  $\tau$  is not identified for Checkerboard; its estimates are kept but flagged.

Table 7: Distribution of block-level mean squared error by design across the 160 blocks under queen contiguity at  $\tau = 1$ , pooled over the five incidence configurations (32 blocks in each). At a fixed value of  $\tau$ , each block contributes one scenario per design. Rows are sorted by median, which here coincides with the order of pooled mean MSE under queen contiguity (Table 2).

| Design                                | Best  | Q25   | Median | Q75   | Worst |
|---------------------------------------|-------|-------|--------|-------|-------|
| Incidence-Guided Saturation Quadrants | 0.037 | 0.045 | 0.080  | 0.135 | 0.169 |
| Balanced Quartiles                    | 0.034 | 0.042 | 0.115  | 0.221 | 0.290 |
| Isolation Buffer                      | 0.128 | 0.150 | 0.159  | 0.169 | 0.206 |
| High Incidence Focus                  | 0.139 | 0.160 | 0.234  | 0.322 | 0.424 |
| 2×2 Blocking                          | 0.048 | 0.062 | 0.271  | 0.580 | 0.719 |
| Checkerboard                          | 0.051 | 0.063 | 0.899  | 2.093 | 2.843 |

neighbor types, in every incidence configuration, at every value of  $\tau$ , and under the control-only regime; its worst case across blocks was the lowest of the six designs; and its intervals had coverage of 0.937 under queen contiguity and 0.942 under rook contiguity. Balanced Quartiles also performed well. It was second overall under both neighbor types and was marginally better than Incidence-Guided Saturation Quadrants only under the both-arms regime (0.0421 against 0.0459 under queen contiguity, 0.0421 against 0.0451 under rook contiguity), where Incidence-Guided Saturation Quadrants had the lower MSE in 22 of 80 queen blocks and 23 of 80 rook blocks. Under the control-only regime, Incidence-Guided Saturation Quadrants had the lower MSE in all 80 queen blocks and 75 of 80 rook blocks, by a wider margin (0.135 against 0.219 under queen contiguity).

Ordering the saturations by incidence made no detectable difference to Incidence-Guided Saturation Quadrants: Saturation Quadrants, which assigns the same four saturations to the quadrants in random order, could not be distinguished from it under either neighbor type (Methods). Coverage did not separate the designs: it was close to 0.94 for every design under both neighbor types, except Checkerboard under rook contiguity, where  $\tau$  is not identified.

## **Application: A Planned Sudden Unexpected Death Prevention**

### **Pilot in North Carolina**

#### **Background**

SUD in adults is a substantial and incompletely understood cause of premature mortality. The SUDDEN (Sudden Unexpected Death in North Carolina) methodology, a validated algorithm for classifying deaths as sudden and unexpected from death certificate and medical record data, was developed and screened against expert physician panel review by Nanavati and colleagues [31]. Using this methodology, Mirzaei and colleagues estimated a North Carolina SUD base rate of approximately 35 per 100,000 among adults aged 20–64 and found that the resulting years of life lost and productivity loss from SUD exceed that of any individual cancer, underscoring SUD’s substantial and under-recognized public health burden [24]. Subsequent work has documented substantial geographic and socioeconomic gradients in SUD incidence within North Carolina specifically: Gan and colleagues found elevated SUD incidence in rural counties relative to urban counties [32], and Mounsey and colleagues found an inverse relationship between household income and SUD incidence in Wake County [33]. Together, this literature establishes both that SUD is a meaningful and measurable outcome in North Carolina and that its incidence varies spatially and socioeconomically in ways that are relevant to the incidence-informed designs evaluated in this chapter.

#### **The planned pilot design**

A pilot intervention is being planned in collaboration with North Carolina’s community college (CC) system, which provides the spatial and organizational unit for the design framework developed in this chapter. North Carolina’s 58 community colleges collectively serve all 100 counties in the state; each college has a primary campus in one county and, in many cases, satellite campuses in neighboring counties (for example, Durham Technical Community College

operates a satellite campus in Orange County in addition to its primary Durham County campus). This network structure motivates the spillover term in this chapter’s outcome model. Community colleges within a shared service network communicate and coordinate programmatic activity with one another, so an intervention delivered at one college plausibly influences outcomes in a neighboring college’s service area even without any formal expansion of the intervention itself. The spillover term in Equation 2 approximates this mechanism: it passes spillover between adjacent clusters through the contiguity weights  $W$ , whereas coordination between colleges follows their programmatic ties, which need not coincide with shared service-area boundaries.

The planned pilot site is Lenoir Community College, whose service area comprises Lenoir, Greene, and Jones counties; the pilot cluster is that whole service area, with anticipated spillover to adjacent CC service areas; the pilot intervention involves workforce development and disaster-response-style training intended to empower CC students and staff to recognize signs of medical distress, connect at-risk individuals to care, and support timely follow-up. Because the teaching intervention under consideration is delivered county-wide by design, rather than at finer spatial resolution such as the census tract level, the treatment unit for this application is the CC service area (equivalently, a group of one or more counties) rather than the census tract, notwithstanding that individual-level incidence data may still be collected at finer resolution to inform the incidence-aware design components. The eventual evaluation plan anticipates that the intervention’s effect will be visible first in intermediate process measures, such as medication adherence and the timeliness of clinic visits following an at-risk event, since meaningful shifts in the SUD incidence rate itself are not expected to be detectable on the timescale of an initial pilot; this expected measurement lag between process outcomes and the ultimate incidence outcome modeled throughout this chapter is discussed further as a limitation below.

County-level SUD incidence for 2018–2021 was derived by Habib from North Carolina death certificates [34]. Of the 412,514 deaths recorded statewide over that period, that derivation kept

the 111,665 deaths among working-age adults (aged 18–64). It excluded deaths in hospitals and other institutions and deaths from expected causes (such as trauma, overdose, COVID-19, and end-stage chronic disease), and it applied a death-certificate algorithm validated against the SUDDEN physician-adjudicated cohort [32, 31]. The result is 21,147 sudden unexpected out-of-hospital deaths among working-age adults, pooled to the county level using death-certificate zip codes, with population-at-risk denominators from Surveillance, Epidemiology, and End Results (SEER) data for the corresponding years. We then aggregate these county-level data to the 58 community college service-area clusters. These data are not yet available for use in this dissertation chapter: access to them requires joining an active institutional review board protocol governing the SUDDEN dataset, a process that is underway at the time of this writing. Planned covariates for the eventual real-data application, once finalized, include county-level median income, food-desert status, physician access, and average driving distance to the nearest hospital, none of which are incorporated into the placeholder illustration below.

## **A preliminary, explicitly placeholder illustration**

As a preliminary illustration only, and not as a finalized design recommendation, we re-ran the six-design comparison pipeline over the actual 58-cluster CC service-area geography, using dissolved county-level service boundaries and Queen contiguity, rather than the abstract  $10 \times 10$  regular grid used in the primary comparison above. Because real SUD incidence data is not yet available, the baseline incidence surface used in this application run is a synthetic placeholder, generated from a Poisson spatial autoregressive process calibrated to the same 35-per-100,000 base rate reported above, with population drawn from 2024 North Carolina Office of State Budget and Management county population estimates. **The numeric results reported in this subsection are a simulated placeholder and must not be interpreted as a design recommendation for the real application;** they will be superseded once the real SUDDEN-derived county dataset described above is finalized and substituted for the synthetic

incidence surface.

With that caveat, the qualitative pattern observed on this real, irregular geography was consistent with the primary grid-based comparison reported above: designs that stratify or saturate treatment according to incidence (Balanced Quartiles and the saturation-family designs) again achieved the lowest average MSE, in the range of 0.075–0.078, while a design adapted from the grid-based Checkerboard concept and a design that ignores incidence entirely (High Incidence Focus, applied without a reliable real incidence signal) performed worse, with average MSE above 0.23. This qualitative consistency across a substantially different, irregular real-world geography is evidence that the primary comparison’s conclusions are not an artifact specific to the regular  $10 \times 10$  grid, though it is not a substitute for repeating the full comparison once real incidence data replace the synthetic placeholder.

## Discussion

### Summary and relevance

We compared six treatment assignment designs for a spatial cluster randomized trial on a grid of 100 clusters, under three mechanisms of heterogeneous baseline incidence, two spillover regimes, and a range of values for spatial dependence, spillover magnitude, and the treatment effect. Some findings held across the incidence configurations and the values of  $\rho$ ,  $\gamma$ , and  $\tau$  under queen contiguity. Others depended on the spillover regime, which a trial team is unlikely to know when it chooses a design.

The robust findings are these. Incidence-Guided Saturation Quadrants had the lowest mean squared error under both neighbor types (0.091 under queen contiguity and 0.072 under rook contiguity), and Balanced Quartiles was second (0.131 and 0.084). Under queen contiguity, the order of the six designs was the same in each of the five incidence configurations and at every value of  $\tau$ , and neither the strength of spatial dependence in the outcome nor the spillover

579 magnitude changed it. The mechanism that generated the incidence surface changed each  
580 design’s MSE only slightly. Coverage of the 95% intervals was close to 0.94 for every design  
581 under both neighbor types, except Checkerboard under rook contiguity, so the designs differed  
582 in precision and coverage did not separate them.

583 The fragile findings concern the spillover regime. Under queen contiguity, Checkerboard and  
584  $2 \times 2$  Blocking were close to the best designs when spillover reached both arms and were the  
585 two worst when it reached only control clusters: Checkerboard’s MSE was 0.062 under the  
586 both-arms regime and 2.111 under the control-only regime, and that of  $2 \times 2$  Blocking was 0.062  
587 and 0.576. Balanced Quartiles also lost ground under the control-only regime (0.042 and 0.219),  
588 while Incidence-Guided Saturation Quadrants was first or second under both regimes. A design  
589 whose performance depends on the regime carries a risk that its average MSE does not show.

590 Checkerboard failed for different reasons under the two neighbor types. Under queen contiguity,  
591 its high MSE was variance, not bias. Each interior cluster has four treated and four control  
592 neighbors, so  $WZ = 1/2$  there, and under the control-only regime the spillover covariate is a  
593 linear combination of the intercept and  $Z$  at 64 of the 100 clusters. Only the boundary clusters  
594 separate the two effects. The estimates stayed centered on  $\tau$  (bias 0.004 under the control-only  
595 regime) and their intervals kept close to nominal coverage (0.942), but they were imprecise.  
596 Under rook contiguity,  $\tau$  is not identified (Methods): the estimates were centered on  $\tau - \gamma$ , so  
597 rook Checkerboard’s MSE grew with  $\gamma$ , and its coverage was 0.150. Checkerboard intersperses  
598 treated and control clusters as finely as the grid allows; both failures come from the dependence  
599 this pattern creates between  $Z$  and  $WZ$ .

600 We recommend Incidence-Guided Saturation Quadrants for the reasons given in the Results,  
601 chiefly its lead under the control-only regime and its lowest worst-case MSE across blocks.  
602 Balanced Quartiles also worked well; its advantage under the both-arms regime was small, and  
603 its disadvantage under the control-only regime was large.

The saturation design did as well when its saturations were assigned without reference to incidence (Saturation Quadrants; see Results). Both designs treat the four quadrants at saturations from 20% to 80%, but no design in this chapter removes that structure while keeping the rest fixed, so these results do not show that the saturation structure is the reason for their lower MSE.

These are design-stage results, and they complement analysis-stage modeling rather than replace it. Every design was analyzed with a spatial lag model that included the spillover term, and for every design whose  $\tau$  was identified the estimates were close to unbiased: the absolute bias of each design under each regime was at most 0.048 under queen contiguity and 0.027 under rook contiguity. Given that model, the choice of design determined how precisely  $\tau$  was estimated.

## Relation to Chapter 2

Checkerboard is the strict allocation of the block-stratified design that Chapter 2 recommended over simple random assignment on rook grids of 8 to 12 clusters. Chapter 2 used the term “checkerboard paradox” for the amplified bias it observed for this design on its  $3 \times 4$  grid when spillover reached both arms and the outcome was spatially dependent. In its limitations, it also noted that an allocation in which treatment and spillover exposure are exactly collinear gives  $\hat{\beta} \approx \beta - \psi$  in its notation. A rook checkerboard is such an allocation on a rectangular grid of any size, and on 100 clusters its coefficient on  $Z$  estimated  $\tau - \gamma$  in the same way. Under queen contiguity, a neighbor structure that Chapter 2 listed as an extension,  $\tau$  is identified, and Checkerboard performed close to the best designs under the both-arms regime and worst under the control-only regime. Chapter 2 compared the design only with simple random assignment; against the wider set of designs studied here, Checkerboard had the highest mean MSE under both neighbor types.

## Limitations

Several limitations bound these conclusions.

First, the primary estimator is an oracle. It includes the exact spillover exposure term, so its results are a best case for each design. We fit a non-oracle model that omits the spillover term as a sensitivity analysis. At  $\tau = 1$ , setting aside Checkerboard under rook contiguity, it was biased for every design, with bias between  $-0.12$  and  $-0.40$  and coverage between  $0.51$  and  $0.90$ , and Incidence-Guided Saturation Quadrants still had the lowest MSE under both neighbor types. The non-oracle model is only one departure from the oracle. We did not study an analyst who includes a spillover term of the wrong form, such as one built on the wrong neighbor definition or a distance decay.

Second, the designs and the fitted model used the incidence surface that generated the outcome (the matched-surface assumption in Methods). In practice, planning incidence is estimated from an earlier period and measured with error. The designs that assign treatment by incidence rank, and the covariate adjustment, would then use a noisy surface, and we did not study how that changes the comparison.

Third, in the Poisson mechanism every cluster had the same population of 100,000. Real clusters, including the community college service areas in the Application, differ in population, so their observed rates differ in precision, and unequal populations may change how the designs that use incidence perform.

Fourth, all results come from a single regular  $10 \times 10$  grid, with  $\beta$  and  $\sigma$  fixed at 1. We did not vary the grid's size or shape, and some results depend on its geometry, including the number of clusters Isolation Buffer treats and the 36 boundary clusters that identify  $\tau$  for Checkerboard under queen contiguity.

Fifth, the simulation shares random draws by design. Blocks within an incidence configuration share its ten incidence surfaces, and the five values of  $\tau$  reuse the same assignments and noise. Because of the shared surfaces, the block-level tests treat dependent blocks as independent, which is why we also compared adjacent designs over the 50 independent configuration-by-surface

units. Because of the shared draws across  $\tau$ , the stable design order under queen contiguity shows only that the order did not change across the five values of  $\tau$  we tested, given the same draws (under rook contiguity,  $2 \times 2$  Blocking and Isolation Buffer swapped places at one value of  $\tau$  under each ordering); it is not an independent replication at each value of  $\tau$ .

Sixth, the simulation’s base rate of 35 per 100,000 is not the rate of the county data described in the Application (see Heterogeneous baseline incidence). We kept it deliberately, because the simulation is a general design study, not a calibration to one data set. In the Poisson mechanism, incidence enters the designs and the fitted model only through the ranks of the observed rates, so the base rate affects the simulation only through the spatial pattern of those ranks and the number of tied rates.

Seventh, in the planned application, changes in SUD incidence are not expected to be detectable on the timescale of an initial pilot, so its evaluation will rely on intermediate process outcomes. We evaluated the designs on a continuous cluster-level outcome generated by the model in Equation 1 and did not study how their performance carries over to a process outcome.

## **Future work**

Four extensions follow from these limitations. First, a data-generating process in which the planning incidence is a noisy snapshot of a latent risk surface, and the outcome depends on the latent surface, would test the matched-surface assumption directly and show how the designs that assign treatment by incidence rank behave when those ranks are partly wrong. Second, unequal cluster populations in the Poisson mechanism would bring the simulation closer to the Application. Third, varying the size and shape of the grid, including irregular geographies such as the community college service areas, would show how much of the ranking depends on a regular  $10 \times 10$  lattice. Fourth, and most immediately, applying the designs to the county-level SUD data described in the Application, once access to those data is in place, will show whether the recommendation holds under observed incidence.

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